

ShanghaiTech University
School of Information Science and Technology

Analog Circuit Lab Experiments

Experiment: MOS Characterization

MOSFET Characterization

Plot I_D vs. V_{DS} curves with different V_{GS} . Label the cutoff, triode and saturation regions on the plot. What is the channel length modulation λ : _____.

What is the transconductance G_m with a bias of $V_{GS} = 2.2$ V and $V_{DS} = 1.5$ V: _____.

What is the transconductance G_m with a bias of $V_{GS} = 2.2$ V and $V_{DS} = 0.06$ V: _____.

Plot $I_D^{0.5}$ vs. V_G , extract V_{TH} : _____ and K_n : _____.